

MARY SEMERDJIAN

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OBJECTIVE

Motivated and detail-oriented cybersecurity professional with a Master's, Bachelor's, and Associate of Science in Computer Science. Experienced in information security, network security, and data analysis. Seeking an entry-level cybersecurity position to apply technical knowledge and problem-solving skills in protecting organizational assets.

EDUCATION

Master of Science in Computer Science | California State University, Los Angeles | Graduation: May 2024 | GPA 3.9

Bachelor of Science in Computer Science | California State University, Los Angeles | Graduation: May 2022 | Awards: Dean's List at CSULA

Associate of Arts Science in Computer Science, Computer Information Systems, Liberal Arts - Natural Sciences and Mathematics, Liberal Arts - Arts and Humanities, Associate of Science in Physics for Transfer | Los Angeles City College | Graduation: June 2019 | Awards: Dean's List at LACC

WORK EXPERIENCE

NETWORK SECURITY GRADUATE ASSISTANT

California State University Los Angeles, LA, CA

08/2023 – 12/2023

- Supported a graduate-level course, Computer and Network Security class at California State University Los Angeles, with 60+ graduate students, providing assistance with course material and focusing on networking topics - network security tool; assisted a cohort of graduate students in installing and configuring intrusion detection systems, firewall management, and VPNs and enhancing student engagement.
- Collaborating with the supervisor with additional support and contributing to academic schooling through laboratory work, special projects, and various educational activities; providing students with positive feedback, grading their assignments, and improving the student's understanding of network security and complex cybersecurity concepts.

MASTER THESIS

California State University Los Angeles, LA, CA

08/2023 – 05/2024

- Authored a Master's Thesis under the supervision of Dr. Ye, focusing on research related to Distributed Denial of Service (DDoS) attacks during my final year as a graduate student.
- Conducted an analysis of detection techniques and mitigation strategies for Distributed Denial of Service (DDoS) attacks by leveraging machine learning, statistical algorithms, probability theory, and neural networks to effectively analyze and train datasets.
- Acquired experience and developed expertise in machine learning, machine learning frameworks, cybersecurity, and networking, utilizing both small and large datasets, as well as employing GPUs for testing and training the data.

SENIOR CAPSTONE PROJECT | DR. MATHIAS BRIEU (BIOMEDICAL)

California State University Los Angeles, LA, CA

08/2021 – 05/2022

- Learned professional skills in 3D modeling using 3D Slicer and 3D visualization software and experimented with Nvidia AIAA, an API that enables the creation of 3D model objects from trained data, improving automation.

- Focused on creating a custom AI model and training it with our data – MRI images of pelvic organs—utilizing the Clara Train SDK for model training.

PROJECTS

HONEYPOT SERVER

- Configured a Honeypot server to monitor malicious activity in Kali Linux and Ubuntu, attaining insight into potential attacks as well as analyzing network vulnerabilities.

HISTOPATHOLOGIC CANCER DETECTION

- Tested and trained different Machine Learning algorithms to develop an accurate model for classifying images containing either metastatic cancer cells or non-metastatic cancer cells.

DATA SCIENCE FOR BUSINESS ANALYTICS: INTRO TO GOOGLE ANALYTICS, ADOBE ANALYTICS, MICROSOFT POWER BI

- Involved in collecting and analyzing data from multiple sources, applying statistical techniques, and utilizing data visualization tools and dashboards to communicate insights and inform strategy.
- Leveraged data and communicate findings that drive better decision-making in large and small organizations.

MARIADB

- Explored the integration of MariaDB into a Node.js application, exemplified through a book search system example.
- Illustrated the practical use of MariaDB, enfoldng the additional functionalities, including connecting to the MariaDB, querying, inserting, updating, and deleting data from MariaDB.

ARTIFICIAL INTELLIGENCE – GENERATIVE AGENT

- Reproduce generative agents, a form of Artificial Intelligence (AI) that can simulate human-like behavior based on existing data; where the approach originates from the research detailed in the” Generative Agents: Interactive Simulacra of Human Behavior” paper.

TECHNICAL SKILLS

- Operating Systems: Windows, Kali Linux, Mac OSX, Ubuntu
- Programming Languages: C++, C, Java, JavaScript, Python, HTML/CSS, PHP, SQL, Haskell, Kotlin, BASIC, Swift
- Networking and Security Tools: Cisco, Nmap, Metasploit, Snort, Wireshark, Nessus
- Data and Medical Visualization Tool: Jupyter Notebook and 3D Slicer
- Machine Learning Tools: Scikit-learn, Neural Networks, and Keras
- Hardware: NVIDIA GeForce RTX
- Database Technologies: MySQL, MariaDB, MongoDB, phpMyAdmin
- Graphic Design Software: Adobe Photoshop
- Web Development Frameworks: Bootstrap